

Mid Artificial Intelligence & Machine Learning

Q1: Choose The Correct Answer

1. Artificial Intelligence is defined as ?

- a. *The technical science that studies and develops theories, method, and applications for simulating and extending human intelligence.*
- b. *The process where computers simulate or implement human learning behavior to acquire new knowledge or skills.*
- c. *The study of the artificial neural networks that contain multiple layers, such as Multi-Layer Perceptron.*
- d. *The study of the biology of the human nervous system.*

2. Training an artificial intelligent system using input vector and supplying the target output is referred to as ?

- a. *Competitive learning.*
- b. *Self-Organized learning.*
- c. *Supervised learning.*
- d. *Unsupervised learning.*

Consider the confusion matrix shown in Table Q1, Answer 3,4,5:

3. Sensitivity ?

- a. 60.5%
- b. 70.0%
- c. *46.0%*
- d. 38.0%

4. Precision ?

- a. 60.5%
- b. 70.0%
- c. *46.0%*
- d. 38.0%

5. Accuracy ?

- a. 60.5%
- b. 70.0%
- c. *46.0%*
- d. 38.0%

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6. Linear regression is ?
 - a. Optimization learning method
 - b. **Statistical learning method**
 - c. Unsupervised learning method
 - d. Correlation-based learning method
7. (_____) occurs when the training error is too large ?
 - a. Reinforcement
 - b. Overfitting
 - c. **Underfitting**
 - d. None of them
8. The XOR gate can be built by using (_____) ANN ?
 - a. Simple perceptron
 - b. One layer
 - c. **Backpropagation**
 - d. Linear regression
9. A learning process in which the system forms natural groupings of the input patterns ?
 - a. **Unsupervised learning**
 - b. Reinforcement learning
 - c. Supervised learning
 - d. Linear regression
10. What is Backpropagation learning algorithms ?
 - a. It's transfer of error back through the network to adjust the input.
 - b. **It's transfer of error back through the network to allow weights to be adjusted.**
 - c. It's transfer of error back through the network using a set of recurrent connections.
 - d. It's the transfer of output back from the hidden layer to the input layer using a set of recurrent connections.
11. What is gradient descent ?
 - a. **Method to find the absolute minimum of a function.**
 - b. Maximum or minimum depends on the situation.
 - c. Method to find the absolute maximum of a function.
 - d. Method to find the average of a function.

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12. Which of the following statements is the best description of the principle of the Hebbian network ?

- a. If two neurons at either end of a weighted link are activated at different times, then increase the value of the link weight.
- b. If two neurons at either end of a weighted link are activated simultaneously, then decrease the value of the link weight.
- c. If two neurons at either end of a weighted link are activated simultaneously, then increase the value of the link weight.
- d. If two neurons at either end of a weighted link are activated, then their weight link will equal zero.

13. Pattern classification problem can be solved by ?

- a. Unsupervised learning
- b. Competitive learning
- c. Supervised learning
- d. Traditional algebra equations

14. What is Neural Network generalization ?

- a. The ability of neural networks to approximate the desired output values for pattern vectors that aren't in the test set.
- b. The ability of neural networks to approximate the desired output values for pattern vectors that aren't in the training set.
- c. The ability of neural networks to generalize on pattern vectors that aren't in the training set.
- d. The ability of neural networks to generalize on pattern vectors that aren't in the test set.

15. Function approximation may be performed by () process ?

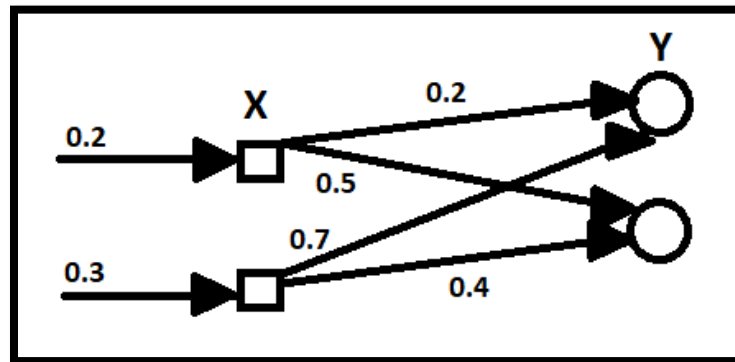
- a. Unsupervised learning
- b. Competitive learning
- c. Supervised learning
- d. Traditional algebra equations

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Q2: For the Kohonen model shown in Figure and using the Euclidean distance.

- find the winning neuron and apply weight updates, Assume learning rate $(\alpha)=0.2$.
- Differentiate between the Hebbian and Kohonen learning methods.



Solution : (a)

$$d_1 = \sqrt{(0.2 - 0.2)^2 + (0.3 - 0.7)^2} = 0.4$$

$$d_2 = \sqrt{(0.2 - 0.5)^2 + (0.3 - 0.4)^2} = 0.316$$

$$d_2 < d_1 ; \text{ } d_2 \text{ is winner}$$

$$\Delta w_{12} = 0.2 (0.2 - 0.5) = -0.06$$

$$\Delta w_{22} = 0.2 (0.3 - 0.4) = -0.02$$

$$w_{12} = 0.5 - 0.06 = 0.44$$

$$w_{22} = 0.4 - 0.02 = 0.38$$

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Solution : (b)

Hebbian :

- 1. If two neurons on either side of a connection are activated synchronously, then the weight of that connection is increased.**
- 2. If two neurons on either side of a connection are activated asynchronously, then the weight of that connection is decreased.**

Kohonen :

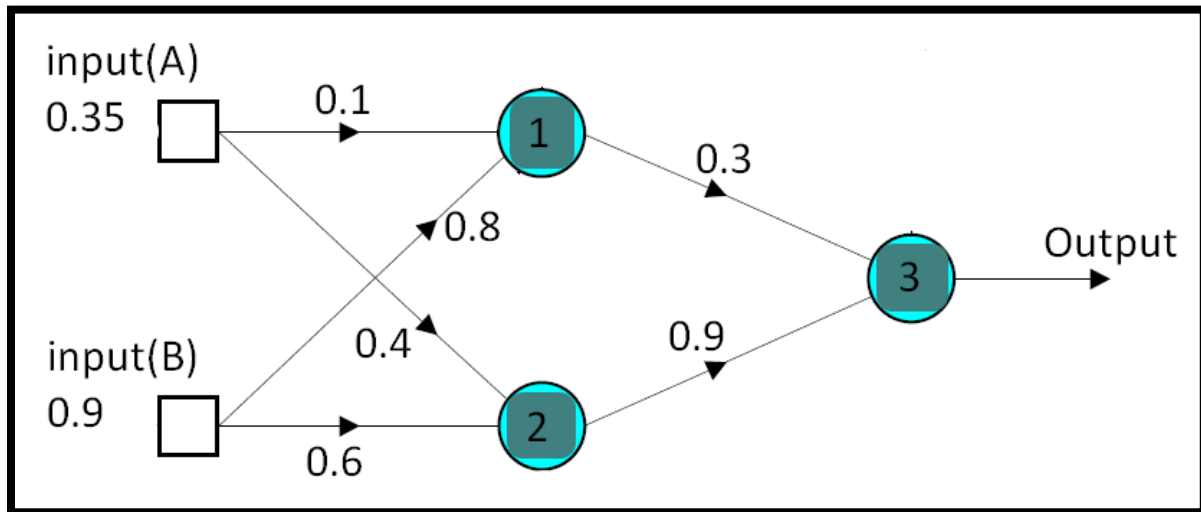
- The Kohonen model provides a topological mapping. It places a fixed number of input patterns from the input layer into a higher-dimensional output or Kohonen layer.
- Training in the Kohonen network begins with the winner's neighbourhood of a fairly large size. Then, as training proceeds, the neighbourhood size gradually decreases.

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Q3: Consider the neural network shown in Figure, Simulate the training process of this network by finding ?

- The output of the network.
- The weight update between neuron 1 and neuron 3.
- The weight update between input B and neuron 2.



Solution : (a)

$$\begin{aligned} Y_1 &= \text{sigmoid} [(0.1 * 0.35) + (0.8 * 0.9) - 0] \\ &= \frac{1}{1 + e^{-x}} = 0.680 \\ Y_2 &= \text{sigmoid} [(0.4 * 0.35) + (0.6 * 0.9) - 0] \\ &= \frac{1}{1 + e^{-x}} = 0.663 \\ Y_3 &= \text{sigmoid} [(0.680 * 0.3) + (0.663 * 0.9) - 0] \\ &= \frac{1}{1 + e^{-x}} = 0.690 \end{aligned}$$

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Solution : (b,c)

$$\begin{aligned} e &= Y_d - Y_3 & \delta_3 &= Y_3(1-Y_3)e \\ &= 0.5 - 0.690 & &= -0.0406 \\ &= -0.19 & & \end{aligned}$$

$$W_{13} = W_{13P} + \Delta W_{13}$$

$$\begin{aligned} \Delta W_{13} &= \alpha * Y_{31} * \delta_3 \\ &= -0.0276 \\ &+ 0.3 \quad \boxed{= 0.272} \end{aligned}$$

$$\begin{aligned} \delta_2 &= Y_2(1-Y_2) \cdot \delta_3 \cdot w_{23} \\ &= 0.663(1-0.663) \cdot -0.0406 \cdot 0.9 \\ &= -0.00816 \end{aligned}$$

$$\begin{aligned} \Delta W_{B2} &= \alpha * B * \delta_2 \\ &= 1 * 0.9 * -0.00816 \\ &= -0.00734 \end{aligned}$$

$$W_{B2} = -0.00734 + 0.6 = \boxed{0.592}$$

- Assume that the neurons have a Sigmoid activation function and learning rate $\alpha = 1$ and target (desired output) = 0.5 and Θ for all neurons = 0

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